COURSE SYLLABUS

CSC14118 – Introduction to Big Data

1. GENERAL INFORMATION

Course name:	Introduction to Big Data
Course name (in Vietnamese):	Nhập môn Dữ liệu lớn
Course ID:	CSC14118
Knowledge block:	Specialized knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	None
Prior-course:	None
Instructors:	Nguyễn Ngọc Thảo, PhD., and Lê Ngọc Thành, PhD.

2. COURSE DESCRIPTION

The course is designed to provide students with an overview of the actively growing field of Big Data, including the principles of Big Data, the benefits and challenges of large-scale data analytics in modern life, and the vast number of Big Data tools applied in business intelligence and scientific research. Large-scale data has unique characteristics that require scaling up analytical technologies and algorithms, leading to new perspectives in data understanding and analysis. The course mainly guides students on handling data processes in batch mode and streaming mode using Apache Hadoop and Apache Spark. Furthermore, the students will collaborate and share more Big Data topics, helping everyone attain a broad view of Big Data in practice.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Understand the principles of Big Data and the motivation for moving from classical data analytics to large-scale data analytics	
G2	Explain the fundamental considerations in large-scale data management: scalability, parallel processing on distributed storage, and streaming processing	
G3	Manipulate Apache Hadoop, Apache Spark, and MongoDB as introductory tools to large-scale data analytics	
G4	Attain a broad view of Big Data in practice through studying and sharing experiences about applications, tools, and techniques	
G5	Participate in a designated online course about some Big Data topics and complete most learning items in that course	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Explain the characteristics of data in Big Data	I
G1.2	Understand the key considerations in handling large-scale data: benefits and challenges, the need of scalable algorithms and techniques, and applications	T
G2.1	Describe the key components in a Big Data cluster	T
G2.2	Interpret the mechanism of parallel processing on distributed storage	T
G2.3	Differentiate batch mode and streaming mode in data processing	T
G3.1	Install Apache Hadoop and implement simple Hadoop MapReduce programs	T/U
G3.2	Install Apache Spark and implement simple data analytical tasks using MLlib	T/U
G3.3	Set up MongoDB Atlas and use it in conjunction with Python	T/U
G3.4	Perform simple data streaming tasks using Spark Streaming	T/U
G4.1	Recognize the wide availability of Big Data analytics tools and frameworks as well as their applications in academy and industry	I

G4.2	Identify the prominent characteristics of some widely known Big Data analytics tools and frameworks	I
G5.1	Understand the learning materials and complete most questions, assignments, and projects in the designated online course	U
G5.2	Present in summary the knowledge and skills attained from the online course	U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (S: Student, T: Teacher)	Assessment
1	Fundamentals of Big Data	G1.1-2 G4.1-2	**In class** • Lecturing (T) **After class** • Group registration (S)	
2	Hadoop Fundamentals: Basic concepts and Ecosystem	G2.1-2 G3.1 G4.1-2	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Announce the list of topics for seminar (T) • Lecturing and demonstrating Hadoop (T) • Case studies and Discussion (S-T) **After class** • Seminar: Topic registration (S)	LW01
3	Hadoop Distributed Filesystem	G2.2 G3.1	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating HDFS (T)	
4	Hadoop MapReduce	G2.2 G3.1	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating MapReduce (T)	LW02

5	Spark Fundamentals: Basic concepts and Ecosystem	G2.2 G3.2 G4.1-2	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating Spark (T) • Case studies and Discussion (S-T)	
6	Spark APIs	G2.2 G3.2	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating Spark (T) • Case studies and Discussion (S-T)	LW03
7	Advanced Analytics with Spark	G2.2 G3.2	**Before class:** • Read the corresponding materials (S) **In class** • Lecturing and demonstrating Spark (T)	
8	NoSQL databases MongoDB	G2.2 G3.3 G4.1-2	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating MongoDB: MongoDB Atlas and MongoDB Compass (T)	
9	Stream processing Spark Streaming	G2.2-3 G3.4	**Before class:** • Read the corresponding materials (S) • Watch the designated video(s) and complete the assignments (S) **In class** • Lecturing and demonstrating Spark streaming (T) **After class** • Seminar: Submit materials (S)	LW04
10	A roadmap to GenAI tools	G1.1-2 G4.1-2	**Before class:** • Read the corresponding materials (S) **In class** • Lecturing and demosntration (T)	

			After class • Seminar: Submit materials (S)	
11	Seminar	G4.1-2 G5.1-2	**Before class:** • Read the presentation materials (S) **In class** • Give an oral presentation of topics learnt in the online course (S) • Q&A (T)	Oral presentation

LABORATORY

The teaching assistants are responsible for

- Consolidating students' comprehension by giving tutorials in office hours (on demand),
- Organizing review sessions for midterm and/or final examinations, and
- Giving, correcting, and grading take-home assignments.

The lab instructors are responsible for

- Consolidating students' problem-solving and technical skills on frameworks and tools, and
- Organizing Q&A sessions for lab work (on demand), and
- Giving, correcting, and grading lab work.

Students will not have weekly classes for laboratory work. Instead, they will contact TA or lab instructors when necessary.

ID	Topic	Course outcomes	Teaching/Learning Activities	Assessments
1	The Fundamentals of Big Data	G1.1-2 G4.1-2		
2	Hadoop Fundamentals: Basic concepts and Ecosystem	G2.1-2 G3.1 G4.1-2	Examine Hadoop and its configurations, and raise questions about issues (S)	LW01
3	Hadoop Distributed Filesystem	G2.2 G3.1	Examine HDFS and its configurations, and raise questions about issues (S)	

4	Hadoop MapReduce	G2.2 G3.1	Examine MapReduce and its configurations, and raise questions about issues (S)	LW02
5	Spark Fundamentals: Basic concepts and Ecosystem	G2.2 G3.2 G4.1-2	Examine Spark and its configurations, and raise questions about issues (S)	
6	Spark API	G2.2 G3.2 G4.1-2	Examine Spark and its configurations, and raise questions about issues (S)	LW03
7	Advanced Analytics with Spark	G2.2 G3.2	Try Spark APIs on Google Colab, and raise questions about issues (S)	
8	NoSQL databases MongoDB	G2.2 G3.3 G4.1-2	Connect to MongoDB, run simple data processing tasks, and raise questions about issues (S)	
9	Stream processing Spark Streaming	G2.2-3 G3.4	Try Spark Streaming on personal computers and Google Colab, and raise questions about issues (S)	LW04
10	A roadmap to GenAI tools	G1.1-2 G4.1-2		
11	Seminar	G4.1-2 G5.1-2	Self-study activities	

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	**Assignments**	**Group work**		**55%**
A11	Lab work LW01–LW04	Set up frameworks and tools, and code small programs using MapReduce and PySpark	G3.1-4	30%
A12	Seminar	Choose a Big Data topic, study related materials, give oral presentation, and write report.	G4.1-2 G5.1-2	25%
A2	**Exams**	**Individual work**		**45%**
A21	Midterm exam (Theory part)	*90 minutes, in-class written exam, Open-book.* They are on any topics in any lecture covered and any reading material assigned up to the time the exam is administered	G1.1-2 G2.1-2	15%
A23	Final exam (Theory part)	*120 minutes, in-class written exam, Open-book.* They are on any topics in any lecture covered and any reading material assigned up to the time the exam is administered	G1.1-2 G2.1-3 G4.1-2	30%

7. RESOURCES

Textbooks

- Tom White. 2015. **Hadoop: The Definitive Guide** (4th. ed.). O’Reilly Media, Inc.
- Bill Chambers and Matei Zaharia, 2018. **Spark: The Definitive Guide: Big Data Processing Made Simple**. O’Reilly Media, Inc.

Others materials

- Anand Rajaraman and Jef rey David Ulman. 2012. **Mining of Massive Datasets**. Cambridge University Press.
- Dirk deRoos, Paul C. Zikopoulos, Bruce Brown, Rafael Coss, and Roman B. Melnyk. 2012. **Hadoop for Dummies**. John Wiley & Sons, Inc.

Tools

- Docker or VMWare

- Google Colaboratory
- Zoom, Padlet, Google docs, Github, etc. (identified by StudentID)

8. GENERAL REGULATIONS & POLICIES

- Students absent for the mid-term or final exam are considered unqualified for course completion.
- Students must accumulate at least 10% of course credits for lab work.
- Students will have several before-class assignments during the course. There will be a 1% penalty for each assignment missed.
- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.